

INFORMATION PROCESSING THEORY OF MEMORY

EARLY THEORY BY WILLIAM JAMES-

Early interest in a dualistic model of memory began in the late 1800s when James distinguished between immediate memory, which he called primary memory, and indirect memory, which he called secondary memory. James based much of his depiction of the structure of memory on introspection, and he viewed secondary memory as the dark repository of information once experienced but no longer easily accessible.

According to James, primary memory, closely related but not identical to what is now called short-term memory (STM), never left consciousness and gave a faithful rendition of events just perceived.

Secondary memory, or long-term memory (LTM), was conceptualised as paths, etched into the brain tissue of people but with wide individual differences.

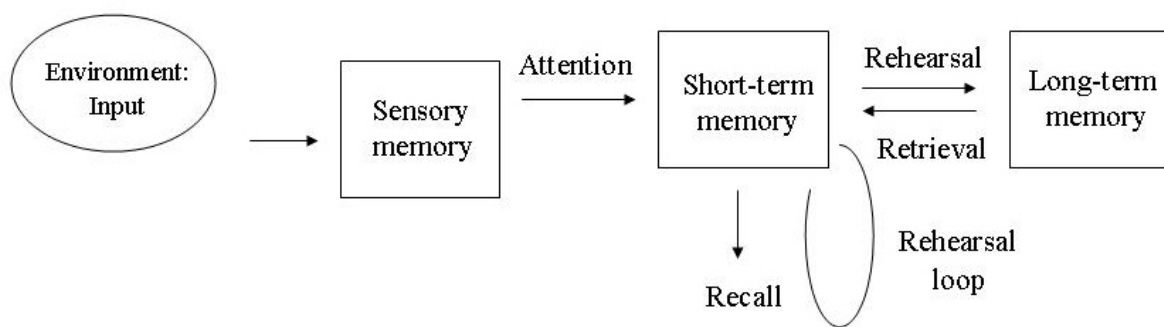
Atkinson and Shiffrin

The proliferation of the boxes in the head explanation for human memory was well underway when Atkinson and Shiffrin (1968) reported their model, the framework of which was based on the notion that memory structures are fixed and control processes are variables. Atkinson and Shiffrin share the dualist concept of memory described by Waugh and Norman but postulate far more subsystems within STM and LTM.

The early models of memory, according to Atkinson and Shiffrin, were too simplistic and not powerful enough to handle the complexities of attention, comparison, retrieval control, transfer from STM to LTM, imagery, coding sensory memory, and so on.

Atkinson and Shiffrin make an important distinction between the concepts of **memory and memory stores**; they use the term “memory” to refer to the data being retained, while “store” refers to the structural component that contains the information.

****Simply indicating how long an item has been retained does not necessarily reveal where it is located in the structure of memory.**



INTRODUCTION-

Traditionally, the most widely used model of information processing is the stage theory model, based on the work of Atkinson and Shiffrin (1968). The key elements of this model are that it views learning and memory as discontinuous and multi-staged. It is hypothesised that as new information is taken in, it is in some way manipulated before it is stored.

❖ SENSORY REGISTER-

Richard Atkinson and Richard Shiffrin (1968)

The sensory store is the initial repository of much information that eventually enters the short- and long-term stores.

In the Atkinson-Shiffrin model, memory starts with a sensory input from the environment. This input is held for a very brief time – several seconds at most – in a sensory register associated with the sensory channels (vision, hearing, touch, and so forth). **This occurs in as little as ½ second for visual stimuli (Sperling, 1960),**

A. ICONIC MEMORY-

The iconic store is a discrete visual sensory register that holds information for very short periods. Its name derives from the fact that information is stored in the form of icons. These in turn are visual images that represent something. Icons usually resemble whatever is being represented.

Sperling's Discovery-1960

In his experiments, Sperling presented the participants with an image of letters displayed in three rows for only 1/20th of a second (50 milliseconds). As the letters disappeared, the participants were asked to report all the letters that they could remember, in a process called Partial report procedure.

X	F	J
R	I	W
L	W	V

B. ECHOIC MEMORY-

Sensory memory of auditory stimuli - 4 or 5 seconds for auditory stimuli (Darwin et al., 1972).

The transfer of new information quickly to the next stage of processing is of critical importance, and sensory memory acts as a portal for all information that is to become part of memory. There are many ways to ensure transfer and many methods for facilitating that transfer. To this end, attention and automaticity are the two major influences on sensory memory, and much work has been done to understand the impact of each on information processing.

❖ STM-SHORT TERM MEMORY-

Although eclipsed in capacity, STM plays an integral role in the memory process.

Information that is attended to and recognised in the sensory register may be passed on to the second stage of information processing, i.e. short-term memory (STM) or working memory, where it is held for perhaps 20 or 30 seconds.

This stage is often viewed as active or conscious memory because it is the part of memory that is being actively processed while new information is being taken in. Some of the information reaching short-term memory is processed by being rehearsed – that is, by having attention focused on it, perhaps by being repeated over and over rehearsal, or by being processed in some other way that will link it up with other information already stored in memory . Generally 5 to 9 units can be processed at any given time in STM.

STM	INFORMATION	CAPCITY	DURATION		
	Visual	7 +_2 items	12 Sec		

	Acuatic				
	Sensory features				
	Semantic information				

CHUNKING-CAPACITY OF MEMORY

George Miller in 1956 emphasises the importance of chunking of information.

No
Chunking

15125551212

Chunking

1 (512) 555 1212

BLOCKING TASK- DURATION OF MEMORY

The Brown-Peterson distractor technique presented a trigram, followed by the directive to count backward from a given number by threes or fours until a light signal prompted the participant to say the trigram. This distractor technique measured short-term recall without the use of rehearsal. When the rehearsal is prevented STM has a narrow capacity.

Displayed:

XJF

481

478

Say Out Loud:

XJF

481

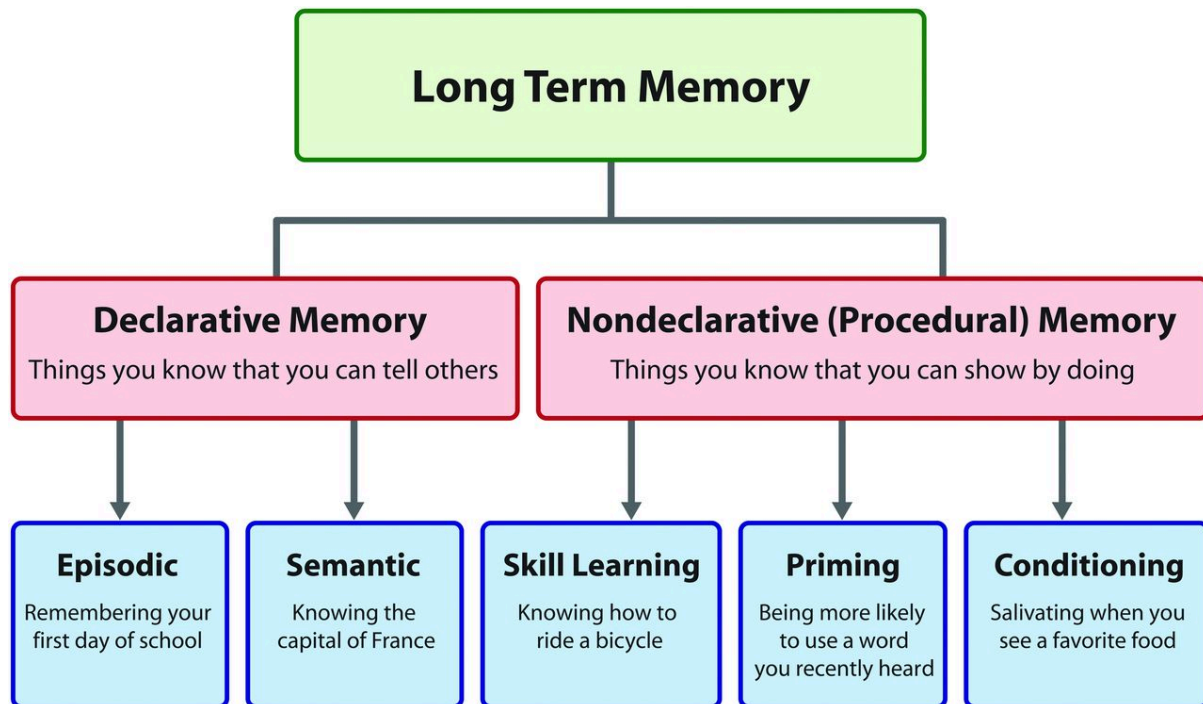
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❖ LONG TERM MEMORY-LTM



1. DECLARATIVE MEMORY

SEMANTIC MEMORY-

Semantic memory is like the knowledge of the world. So, it may include knowledge about culture, history, sports, music, etc. This kind of memory tends to encode, store or organise, retrieve information (knowledge of the world) in the memory according to its meaning. This also talks about how the information is related to other kinds of information that already exists in the memory. There is very little forgetting when information is stored according to the meaning, or logical hierarchies, or semantically (Quillian, 1966). Closely related, but a different type of memory is lexical memory. It is the representational system for the words of our language. Psycholinguists are interested in the structure of lexical memory.

EPISODIC MEMORY-

The memories that are stored with respect to the time when they happened are known as episodic memory. The various episodes of our life experiences that are encoded, stored, retrieved memories in the LTM that are related to our personal individual experiences. They may have a reference to our past events or things that happened to us at particular times.

AUTOBIOGRAPHICAL MEMORY-IS A KIND OF EPISODIC MEMORY.

As the name suggests, it is the memory of your own past events or personal experiences. Our autobiographical memory (AM) is generally accurate but sometimes it is also influenced by the constructive nature of memory. But, do we remember life events from all periods of life equally? Studies have suggested that people from middle age remember life events from their youth period and early-adult period more vividly than their recent past (Read & Connolly, 2007). Marigold Linton (1975, 1982) did a classic study on AM using Ebbinghaus' method of introspection. She kept a diary for six periods, recording at least two events per day. She studied these recorded memories to understand the nature of AM.

PROCEDURAL MEMORY-

Procedural memory is most simply 'knowing how'. Wayne Podrouzek (1994, lecture) gave the example of turning a corner on a bicycle. The angle at which the bicycle must tilt in relation to the speed of the bicycle could be computed mathematically, but that's not how we do it. We just 'know how' automatically. Very simple organisms have this memory system alone (Tulving, 1985). It is a little more advanced than pure instinct since it does involve stimulus response learning, but nothing more.

2.NON DECLARATIVE MEMORY-

Non-declarative memory is expressed by behaviour and does not involve conscious recollection. There are two non-declarative memory systems:

The perceptual representation system – a collection of domain-specific modules that operate on perceptual information about the form and structure of words and objects. Repetition priming occurs within this system. Stimulus processing is faster on the second and successive presentation of a stimulus.

Procedural memory (skill learning) refers to the learning of motor and cognitive skills across a wide range of situations. This type of learning generalises. Perceptual priming occurs when repeated presentation of a stimulus leads to facilitated processing of its perceptual features.

Conceptual priming occurs when repeated presentation of a stimulus leads to facilitated processing of its meaning. Levy et al. (2004) found that amnesic patients and controls showed similar amounts of conceptual priming, but amnesics performed poorly on recognition memory. Insausti et al. (2013) reported a patient who showed perceptual priming but no declarative memory. The notion that perceptual priming depends on different brain systems from those involved in explicit memory is strengthened by finding a double dissociation (Gabrieli et al., 1995).

However, Schacter et al. (1995) found controls showed more priming than did the amnesics when words were presented and tested in the same voice. The implication is that amnesics are poor at binding different kinds of information. Priming is likely to be based on repetition suppression; where the brain activity decreases with stimulus repetition.

Voss et al. (2008) found repetition suppression was associated with reduced activity in the left prefrontal cortex. Foerde and Poldrack (2009) identified types of skill learning or procedural memory: motor skill learning sequence learning mirror tracing perceptual skill learning o mirror reading o probabilistic classification learning o artificial grammar learning. Most tasks involve elements of both declarative and procedural memory. Reber et al. (1996) found that amnesics learn the probabilistic classification task as rapidly as do controls. Foerde et al. (2006) found that learning on the probabilistic classification task can depend on either procedural or declarative memory. Tranel et al. (1994) found that amnesic patients showed comparable learning on the pursuit rotor to controls. Cavaco et al. (2004) found that amnesics showed comparable rates of learning on everyday skill-learning tasks such as weaving and control stick tasks. Declarative and procedural memory are not entirely separate. There can be interactions between the two memory systems. Brown and Robertson (2007) found that procedural learning disrupted learning on a declarative memory task and vice versa.

Non-declarative memory does not involve conscious recollection but reveals itself through behaviour. Repetition priming and procedural memory (skill learning) are two major types of non-declarative memory that may involve separate memory systems (although there is also evidence for a single memory system). Repetition priming can take the form of perceptual or conceptual priming. There is some evidence that different processes underlie the two types of priming. Amnesic patients tend to have preserved skill learning, as well as preserved perceptual and conceptual priming. Priming leads to reduced brain activity, presumably because processing becomes more efficient when stimuli are repeated.

EXTENSION INFORMATION PROCESSING MODEL-

EXTENDED STM MODEL-

In 1974, Baddeley & Hitch, after incorporating the idea of the level of processing (LOP) proposed a new model for STM and termed it as working memory. Thus, working memory can be defined as “a limited-capacity system for temporary storage and manipulation of information for complex tasks such as comprehension, learning, and reasoning” (Goldstein book, pp. 131). Baddeley’s model of Working Memory (WM) consists of four components:

- ★ **central executive,**
- ★ **phonological loop,**
- ★ **visuospatial sketchpad,**
- ★ **episodic buffer**

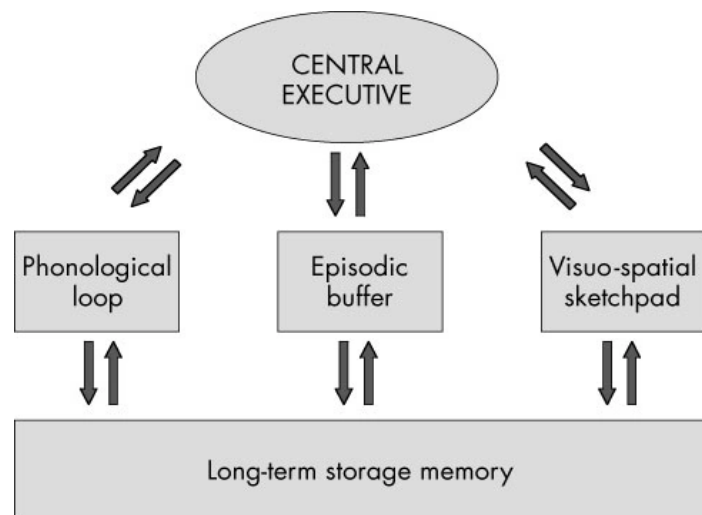
The central executive, as the name suggests, works as an executive in our working memory. It coordinates and regulates cognitive operation between sub-ordinate systems namely, phonological loop, visuospatial, and episodic buffer.

It decides which of the memory will become part of long-term memory and which will fade away.

The phonological loop is responsible for storing verbal and auditory information. The information stored in the phonological loop will decay within 2 seconds unless it is not rehearsed. It consists of two components: the phonological store, which stores information for a few seconds; and the articulatory rehearsal process, responsible for rehearsing the information in order to keep the information stored in the phonological store from decaying. For instance, trying to remember a

phone number, you have just been told by your friend involves the phonological loop.

The visuospatial sketchpad keeps visual and spatial information stored. For instance, the mental picture that comes up in your mind while listening to a story or solving a puzzle involves using your visuospatial



sketchpad.

The episodic buffer is responsible for combining information from the phonological loop, visuospatial, and long-term memory for generating a unitary episodic representation of information. Thus, this component helps us in making a sense of the received information.

CRITICISM-

1. One of the early and central criticisms to the Atkinson-Shiffrin model was the inclusion of the sensory registers as part of memory. Specifically, the original model seemed to describe the sensory registers as both a structure and a control process (Feldman et al 1985).

2. Information processing theory considered STM only as a short-term memory storehouse but later studies disapproved of it (1974, Baddeley & Hitch).
3. The model has been further criticised as suggesting that rehearsal is the key process which initiates and facilitates transfer of information into LTM. his criticism is somewhat unfounded as Atkinson and Shiffrin clearly state a difference between rehearsal and coding, where coding is akin to elaborative processes which levels-of-processing would call deep-processing. In this light, the levels-of-processing framework could be seen as more of an extension of the Atkinson-Shiffrin model rather than a refutation.
4. Endel Tulving 1972, notes the importance of encoding specificity in long-term memory. To clarify, there are definite differences in the way information is stored depending on whether it is episodic (memories of events), procedural (knowledge of how to do things), or semantic (general knowledge).
5. According to the connectionist model parallel distributed processing model , memory forms due to the connection between neural nodes (Fren et al 2003).
6. According to the model of the levels of processing (LOP), whether the information will be retrieved successfully or not depends on its level of processing. LOP refers to the level at which information has been encoded- Craik and Tulving (1975) have proposed three LOP)